1 Forward kinematics models

1.1 DH model

The DH frames are assigned from the CuRo6R URDF, with frame $\{0\}$ at `base_link`, joint i rotating about z_{i-1}, and $q = 0$ matching the URDF home configuration. We use standard DH transforms:

$$^{i-1}T_i = R_z(\theta_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i). \tag{1}$$

Table 1 gives the resulting parameters and the geometric choices that determine them. The only fixed joint-angle offset is $\theta_3 - q_3 = \pi/2$. The full sequence of frame drawings is given in Appendix A.

Table 1: DH parameters of CuRo6R. Lengths are in metres; angular constants are in radians.

i	d_i	θ_i	α_i	a_i	Geometric origin of the parameters
1	1	q_1	$-\pi/3$	0.8	The two base heights add: $d_1 = 0.5 + 0.5$; the lateral offset is $a_1 = 0.8$. Joint 2's axis is its local y axis, so the 30° URDF roll gives $\alpha_1 = 30° - 90° = -60°$.
2	1.25	q_2	$+\pi/2$	2	The joint-3 offset resolves into 1.25 along z_1 and 2 along x_2. At home $x_2 = x_1$, and the orthogonal joint axes give $\alpha_2 = +90°$.
3	0.75	$q_3 + \pi/2$	$+\pi/2$	0	Moving 0.75 along z_2 reaches the intersection with joint 4's axis: O_3 in step 3 of Appendix A, hence $a_3 = 0$. Choosing $x_3 = y_2$ gives the $+90°$ angle offset and $\alpha_3 = +90°$.
4	2.5	q_4	$-\pi/2$	0	The wrist centre lies $1.5 + 1 = 2.5$ along z_3 from O_3. Axes 4 and 5 intersect there; with $x_4 = x_3$ at home, $\alpha_4 = -90°$.
5	0	q_5	$+\pi/2$	0	Axes 5 and 6 meet at the wrist centre, so $O_5 = O_4$ and $d_5 = a_5 = 0$ [Appendix A, step 5]. Keeping $x_5 = x_4$ gives $\alpha_5 = +90°$.
6	1.5	q_6	0	0	The endpoint is 1.5 along z_5 from the wrist centre. Choosing $z_6 = z_5$ and $x_6 = x_5$ at home gives $a_6 = \alpha_6 = 0$.

Endpoint alignment. The endpoint E is `link_6`. Its origin coincides with O_6, while $x_E = z_6$, $y_E = x_6$ and $z_E = y_6$. The fixed alignment is therefore appended to the DH chain to recover the URDF endpoint pose:

$$^0T_E = \left(\prod_{i=1}^{6} {}^{i-1}T_i\right) {}^6T_E, \qquad {}^6T_E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{2}$$

1.2 Product of exponentials model

The product of exponentials (PoE) model uses the joint axes at the home configuration $q = 0$. For each joint, ω_i is its unit axis direction and r_i is its physical origin. All vectors are expressed in `base_link`; the endpoint is `link_6`, as in the DH model.

Axis directions. The rotation axes were identified visually in a URDF view at home. Their positive directions follow the URDF joint convention. The fixed tilt $\gamma = \pi/6$ gives the sine and cosine components of the tilted axes. In particular, $\omega_2 = R_x(\gamma)e_y$ and $\omega_3 = \omega_5 = R_x(\gamma)e_z$, where e_x, e_y, e_z are the base unit vectors.

Joint origins. Starting at joint 1, each origin was found by adding the relevant robot dimensions to the preceding origin. Each offset must first be expressed in the base frame. With lengths in metres, this gives

$$\begin{aligned} r_1 &= \tfrac{1}{2}e_z, & r_2 &= r_1 + \tfrac{4}{5}e_x + \tfrac{1}{2}e_z, \\ r_3 &= r_2 + 2e_x + \tfrac{5}{4}\omega_2, & r_4 &= r_3 + \tfrac{3}{2}e_x + \tfrac{3}{4}\omega_3, \\ r_5 &= r_4 + e_x, & r_6 &= r_5 + \tfrac{3}{2}e_x. \end{aligned} \tag{3}$$

For example, the offset from joint 2 to joint 3 combines a 2 m step along the base x axis and a 1.25 m step along joint 2's tilted axis. The resulting vectors are listed in Table 2, together with $v_i = -\omega_i \times r_i$.

Table 2: Home screw data. Each vector is written as a row; r_i uses the physical joint origin. Lengths are in metres.

i	ω_i^T	r_i^T	v_i^T
1	$(0, 0, 1)$	$(0, 0, \frac{1}{2})$	$(0, 0, 0)$
2	$(0, \frac{\sqrt{3}}{2}, \frac{1}{2})$	$(\frac{4}{5}, 0, 1)$	$(-\frac{\sqrt{3}}{2}, -\frac{2}{5}, \frac{2\sqrt{3}}{5})$
3	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{14}{5}, \frac{5\sqrt{3}}{8}, \frac{13}{8})$	$(\frac{7}{4}, -\frac{7\sqrt{3}}{5}, -\frac{7}{5})$
4	$(1, 0, 0)$	$(\frac{43}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$
5	$(0, -\frac{1}{2}, \frac{\sqrt{3}}{2})$	$(\frac{53}{10}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(\frac{7}{4}, -\frac{53\sqrt{3}}{20}, -\frac{53}{20})$
6	$(1, 0, 0)$	$(\frac{34}{5}, \frac{5\sqrt{3}-3}{8}, \frac{13+3\sqrt{3}}{8})$	$(0, \frac{13+3\sqrt{3}}{8}, \frac{3-5\sqrt{3}}{8})$

Forward model. Each pair (ω_i, v_i) defines a screw axis and its matrix form:

$$\xi_i = \begin{pmatrix} \omega_i \\ v_i \end{pmatrix}, \qquad \widehat{\xi}_i = \begin{pmatrix} [\omega_i]_\times & v_i \\ 0^T & 0 \end{pmatrix}, \qquad [\omega_i]_\times p = \omega_i \times p. \tag{4}$$

The endpoint has orientation $R_x(\gamma)$ and position r_6 at home. This gives its home pose M and the complete forward kinematics:

$$M = \begin{pmatrix} R_x(\gamma) & r_6 \\ 0^T & 1 \end{pmatrix}, \qquad \boxed{{}^0T_E(q) = e^{\widehat{\xi}_1 q_1} e^{\widehat{\xi}_2 q_2} \cdots e^{\widehat{\xi}_6 q_6} M.} \tag{5}$$

The screw axes stay at their home values; the exponentials are multiplied in joint order.

1.3 Verification

The DH and PoE models were compared at 514 test configurations. The position residual was below 10^{-14} m and the rotation-matrix Frobenius residual was below 10^{-14}.

2 Jacobian and singularities

Construction from the PoE model. The MATLAB PoE routine keeps the intermediate transforms. These give each joint's current unit direction w_i and a point p_i on its axis, both in `base_link`. For a revolute joint, the Jacobian column is

$$ {}^0J_k(:,i) = \begin{pmatrix} w_i \times (p_k - p_i) \\ w_i \end{pmatrix}, \qquad i = 1, \ldots, 6. \tag{6} $$

The upper rows give linear velocity and the lower rows give angular velocity. We use $k = 6$ for the endpoint. Choosing $k = 5$ instead refers the same tool motion to the wrist centre.

Wrist Jacobian. Here ${}^3J_5$ denotes this full Jacobian at the wrist centre, expressed along the DH frame-3 axes. Both velocity blocks are rotated by $({}^0R_3)^T$; the velocities remain relative to the base. All three wrist axes pass through p_5, so their linear columns vanish. With $c_i = \cos q_i$ and $s_i = \sin q_i$, the result has the form

$$ {}^3J_5 = \begin{pmatrix} A & 0 \\ C & B \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -s_4 & c_4 s_5 \\ 0 & c_4 & s_4 s_5 \\ 1 & 0 & c_5 \end{pmatrix}. \tag{7} $$

The last three columns describe the wrist subchain. The blocks A and C give the linear and angular contributions of joints 1–3; B gives the wrist's angular contribution.

Symbolic and numerical routes. The same MATLAB function accepts numerical or symbolic inputs. Numerical inputs evaluate the Jacobian at a chosen configuration. Symbolic inputs give an exact expression, from which the determinant is obtained. We use $J = {}^3J_5$, whose zero block makes the determinant easy to evaluate:

$$ \det J(q) = \sin q_5 \, D(q_2, q_3), \qquad D(q_2, q_3) = -\det A. \tag{8} $$

Changing the expression frame or velocity reference point preserves this determinant, provided all six columns describe the same full tool motion. The wrist factor vanishes at $q_5 = 0$ or π modulo 2π. Axes 4 and 6 then lie on the same line, so one independent rotation is lost. The arm factor depends jointly on q_2 and q_3. Thus the wrist and arm conditions separate, while the arm condition must be studied in the (q_2, q_3) plane, as shown in Figure 1.

A full configuration is regular exactly when both factors are nonzero. Numerically, we monitor the smallest singular value of J. It becomes zero at a singularity, and small values indicate that the Jacobian is close to losing rank.

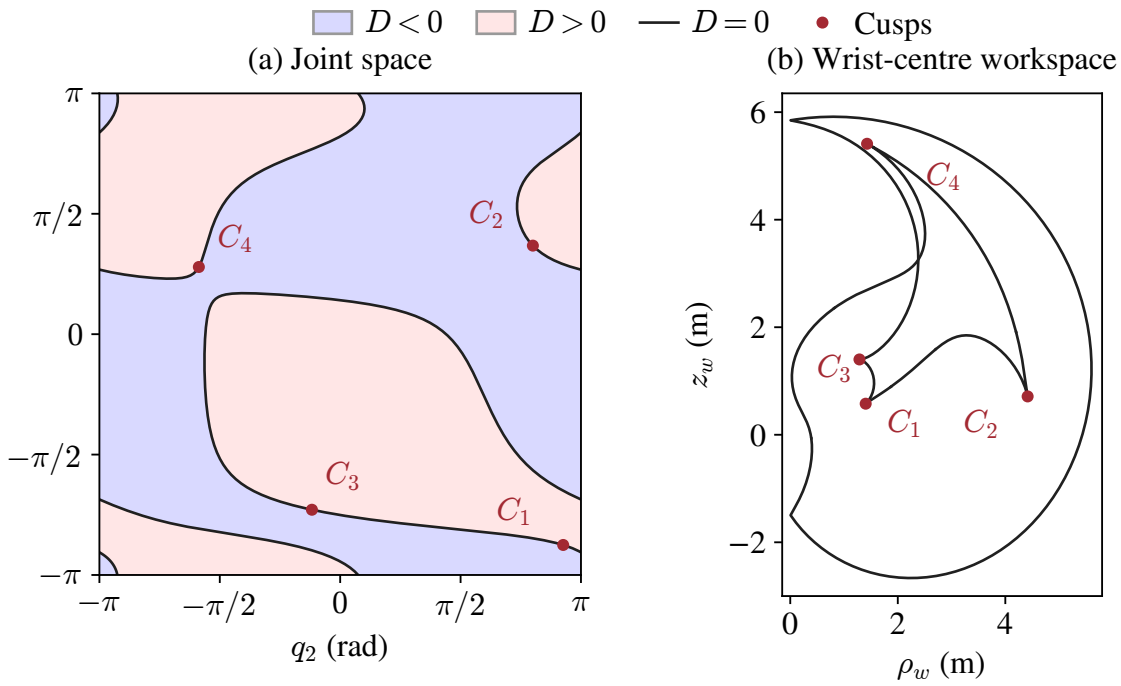


Figure 1: Arm singularities in joint space (left) and the wrist-centre workspace (right), with $\rho_w = \sqrt{x_w^2 + y_w^2}$. Black curves show $D = 0$ and its image under forward kinematics; the colours on the left give the sign of D. Matching labels mark four cusps, where three arm IK solutions coincide. They show that the robot's positioning chain is cuspidal: it can change between IK solutions without crossing an arm singularity. Opposite joint-space edges match modulo 2π; wrist singularities remain a separate condition.

3 Analytical inverse kinematics

The axes of joints 4–6 intersect at the wrist centre, allowing position and orientation to be solved separately. We first determine the arm angles (q_1, q_2, q_3) from the desired wrist-centre position, then recover the wrist angles (q_4, q_5, q_6) from the desired orientation.

3.1 Wrist-centre positioning

Let R_d and p_d denote the desired orientation and position of `link_6` relative to `base_link`. Since the endpoint offset lies along the tool's x axis, the target wrist centre is

$$p_w = p_d - d_6 R_d e_x = (x_w, y_w, z_w)^T, \qquad e_x = (1, 0, 0)^T. \tag{9}$$

Throughout, $c_i = \cos q_i$ and $s_i = \sin q_i$.

Derivation

1. Relate the target wrist position to the three arm angles. For a candidate q_1, undo the base rotation, translate to the shoulder at $s = (a_1, 0, d_1)^T$, and undo the fixed tilt $\gamma = \pi/6$. The resulting auxiliary frame follows the base rotation and fixed tilt, but

not the joint-2 rotation. With $h = z_w - d_1$ and $(c_\gamma, s_\gamma) = (\sqrt{3}/2, 1/2)$,

$$\begin{aligned} r(q_1) &:= R_x(-\gamma)\Big(R_z(-q_1)p_w - s\Big) = (X, Y, Z)^T, \\ X &= c_1 x_w + s_1 y_w - a_1, \\ Y &= c_\gamma(-s_1 x_w + c_1 y_w) + s_\gamma h, \\ Z &= -s_\gamma(-s_1 x_w + c_1 y_w) + c_\gamma h. \end{aligned} \tag{10}$$

These coordinates still depend on the unknown q_1.

In the moving URDF frame of `link_2`, the shoulder-to-elbow vector is $a = (a_2, d_2, 0)^T$. The elbow-to-wrist vector is $R_z(q_3)b_0$, where $b_0 = (d_4, 0, d_3)^T$ is expressed in the URDF frame of `link_3_d3`. Applying the shoulder rotation maps their sum into the auxiliary frame:

$$r(q_1) = R_y(q_2)\Big(a + R_z(q_3)b_0\Big) = R_y(q_2)\begin{pmatrix} a_2 + d_4 c_3 \\ d_2 + d_4 s_3 \\ d_3 \end{pmatrix}. \tag{11}$$

The rotation $R_z(q_3)$ above is expressed using the URDF link frames, with $q_3 = 0$ at the home configuration. The DH frames use a different reference orientation: at home, their angular parameter is already $\theta_3 = \pi/2$. Consequently, $\theta_3 = q_3 + \pi/2$, whereas the rotation in this geometric derivation uses q_3 directly.

2. Express $\sin q_3$ and $\cos q_3$ as functions of q_1. Rotation about y preserves both the y component and the norm. Equation (11) therefore gives two relations independent of q_2:

$$\begin{aligned} Y &= d_2 + d_4 s_3, \\ N := X^2 + Y^2 + Z^2 &= a_2^2 + d_2^2 + d_3^2 + d_4^2 + 2a_2 d_4 c_3 + 2d_2 d_4 s_3. \end{aligned} \tag{12}$$

Setting $K = d_2^2 - a_2^2 - d_3^2 - d_4^2$ yields

$$s_3 = \frac{Y - d_2}{d_4}, \qquad c_3 = \frac{N + K - 2d_2 Y}{2a_2 d_4}. \tag{13}$$

Here $a_2 = 2$ and $d_4 = 2.5$ are nonzero. For a trial value of q_1, these expressions give the required values of $\sin q_3$ and $\cos q_3$. They correspond to a real elbow angle only if $s_3^2 + c_3^2 = 1$. This consistency condition provides the equation used to determine q_1.

3. Determine the candidate base angles q_1. Since $R_x(-\gamma)$ preserves the norm, $N = \|r(q_1)\|^2 = \|R_z(-q_1)p_w - s\|^2$. Defining $N_0 = x_w^2 + y_w^2 + h^2 + a_1^2$ gives the compact expression

$$N = N_0 - 2a_1(x_w c_1 + y_w s_1). \tag{14}$$

Both N and Y are affine in (c_1, s_1). Substitution of (13) into $s_3^2 + c_3^2 = 1$ gives

$$4a_2^2(Y - d_2)^2 + (N + K - 2d_2 Y)^2 = 4a_2^2 d_4^2. \tag{15}$$

Introduce the half-angle variable $t = \tan(q_1/2)$, so that

$$D = 1 + t^2, \qquad c_1 = \frac{1 - t^2}{D}, \qquad s_1 = \frac{2t}{D}. \tag{16}$$

Writing $Y = \widetilde{Y}/D$ and $N = \widetilde{N}/D$, their quadratic numerators are

$$\begin{aligned} \widetilde{Y}(t) &= c_\gamma\Big[y_w(1 - t^2) - 2x_w t\Big] + \tfrac{1}{2}hD, \\ \widetilde{N}(t) &= N_0 D - 2a_1\Big[x_w(1 - t^2) + 2y_w t\Big]. \end{aligned} \tag{17}$$

Multiplying (15) by D^2 gives

$$\boxed{P(t) = 4a_2^2(\widetilde{Y} - d_2 D)^2 + (\widetilde{N} + KD - 2d_2\widetilde{Y})^2 - 4a_2^2 d_4^2 D^2 = 0.} \tag{18}$$

Each quantity being squared is quadratic in t, so P has degree at most four. Every real root supplies a candidate $q_1 = 2\arctan t$. The substitution $t = \tan(q_1/2)$ is undefined at $q_1 = \pi$, so this angle is not represented by any finite root of $P(t)$. To avoid missing a solution, we test $q_1 = \pi$ separately in (15) and retain it only if that equation is satisfied.

4. Recover q_2 and q_3 for each candidate q_1. For each candidate q_1, evaluate (10) and (13), then define $A = a_2 + d_4 c_3$. The remaining components of (11) are

$$X = c_2 A + s_2 d_3, \qquad Z = -s_2 A + c_2 d_3. \tag{19}$$

Thus

$$\begin{aligned} C_2 &= AX + d_3 Z = (A^2 + d_3^2)c_2, \\ S_2 &= d_3 X - AZ = (A^2 + d_3^2)s_2, \end{aligned} \qquad q_2 = \operatorname{atan2}(S_2, C_2). \tag{20}$$

The common factor is positive, so no division is needed before applying atan2. The elbow angle is recovered as $q_3 = \operatorname{atan2}(s_3, c_3)$. Since A uses c_3 directly, q_2 and q_3 can be computed in either order.

If leading polynomial coefficients vanish, the degree reduces. For this robot, $P \equiv 0$ occurs when the target wrist centre is a reachable point on the base rotation axis. At such a position, q_1 is arbitrary, yielding a continuous family of arm configurations.

Practical evaluation

Given (R_d, p_d), compute p_w from (9), followed by h, K and N_0, and form P using (17)–(18). For $P \not\equiv 0$, collect $q_1 = 2\arctan t$ from every real root and test $q_1 = \pi$ separately in (15); for $P \equiv 0$, choose any q_1, for example 0, to represent the continuous family. For each candidate, evaluate (X, Y, Z) from (10), set $N = X^2 + Y^2 + Z^2$, obtain (s_3, c_3) from (13), and set $A = a_2 + d_4 c_3$. Then compute

$$q_2 = \operatorname{atan2}(d_3 X - AZ,\ AX + d_3 Z), \qquad q_3 = \operatorname{atan2}(s_3, c_3).$$

Check each triple against the target wrist position using forward kinematics, represent angles in $[-\pi, \pi]$ and remove duplicates modulo 2π. Pass the valid arm configurations to the wrist-orientation problem.

3.2 Endpoint orientation

For each arm configuration (q_1, q_2, q_3) found in the positioning step, we now determine the wrist angles (q_4, q_5, q_6) that give the desired orientation R_d. Different arm configurations generally require different wrist angles, so this calculation is repeated for each arm solution.

Derivation

1. Express the desired orientation relative to the arm. Using the same URDF link frames as in the positioning derivation, the orientation of `link_3_d3` relative to

`base_link` is

$$R_{\text{arm}} = R_z(q_1)R_x(\gamma)R_y(q_2)R_z(q_3), \qquad \gamma = \pi/6. \tag{21}$$

The wrist joints rotate about their local x, z and x axes, respectively. There are no fixed rotations between these URDF link frames. Thus

$$R_d = R_{\text{arm}}R_{\text{wrist}}, \qquad R_{\text{wrist}} := R_{\text{arm}}^T R_d = R_x(q_4)R_z(q_5)R_x(q_6). \tag{22}$$

Here the transpose undoes the known arm rotation. Let r_{ij} denote the entry in row i and column j of R_{wrist}.

2. Use the first column to recover q_5 and q_4. Since $R_x(q_6)e_x = e_x$, the first column is

$$R_{\text{wrist}}e_x = R_x(q_4)R_z(q_5)e_x = \begin{pmatrix} c_5 & c_4 s_5 & s_4 s_5 \end{pmatrix}^T = \begin{pmatrix} r_{11} & r_{21} & r_{31} \end{pmatrix}^T. \tag{23}$$

Thus $c_5 = r_{11}$. Using $c_4^2 + s_4^2 = 1$ also gives $\rho := \sqrt{r_{21}^2 + r_{31}^2} = |s_5|$. For $\rho > 0$, take each sign $s_5 = \pm\rho$. Dividing the last two entries by s_5 gives c_4 and s_4, so the two angle pairs are

$$q_5 = \operatorname{atan2}(s_5, r_{11}), \qquad q_4 = \operatorname{atan2}\left(\frac{r_{31}}{s_5}, \frac{r_{21}}{s_5}\right). \tag{24}$$

Keep the chosen sign of s_5 in both fractions. The case $\rho = 0$ is treated in step 4.

3. Undo q_4 to recover q_6. Once q_4 is known, multiply on the left by $R_x(-q_4)$ to undo its rotation:

$$B := R_x(-q_4)R_{\text{wrist}} = R_z(q_5)R_x(q_6). \tag{25}$$

The third row of $R_z(q_5)$ is $(0, 0, 1)$, so the third row of B is $(0, s_6, c_6)$. We can therefore recover q_6 without dividing by s_5:

$$q_6 = \operatorname{atan2}(B_{32}, B_{33}). \tag{26}$$

When $\rho > 0$, the two wrist solutions are related by $(q_4, q_5, q_6) \mapsto (q_4 + \pi, -q_5, q_6 + \pi)$, with all angles taken modulo 2π.

4. Handle the cases where $s_5 = 0$. When $\rho = 0$, we have $q_5 = 0$ or $q_5 = \pi$ modulo 2π. The formula for q_4 divides by s_5 and no longer applies. The axes of joints 4 and 6 lie on the same line, so their angles cannot be determined separately. The wrist rotation reduces to $R_x(q_4 + q_6)$ for $q_5 = 0$ and $R_x(q_4 - q_6)R_z(\pi)$ for $q_5 = \pi$. Thus only a sum or a difference is fixed:

$$\begin{aligned} q_5 = 0 \ (r_{11} = 1): &\quad q_4 + q_6 = \sigma := \operatorname{atan2}(r_{32}, r_{22}), \\ q_5 = \pi \ (r_{11} = -1): &\quad q_4 - q_6 = \delta := \operatorname{atan2}(-r_{32}, -r_{22}). \end{aligned} \tag{27}$$

These relations hold modulo 2π. We may choose any q_4 and set $q_6 = \sigma - q_4$ in the first case or $q_6 = q_4 - \delta$ in the second. Choosing $q_4 = 0$ gives the representatives $(0, 0, \sigma)$ and $(0, \pi, -\delta)$ for (q_4, q_5, q_6).

Practical evaluation

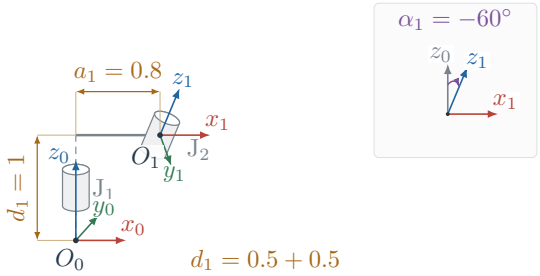
For each arm solution, use both wrist branches from steps 2–3 when ρ exceeds a small numerical tolerance. Otherwise, take $q_4 = 0$ and use the sign of r_{11} to select the case in (27). Combine each wrist solution with its arm solution, represent all angles in $[-\pi, \pi]$, and remove duplicates modulo 2π. Verify the position and orientation against (p_d, R_d) by forward kinematics, using separate tolerances.

A DH frame constructions

Figure 2 collects all six frame-to-frame constructions at the home configuration ($q = 0$), using one fixed viewing direction and length scale. Lengths are in metres; angles are shown in degrees for readability.

1 Base to shoulder: $\{0\} \to \{1\}$

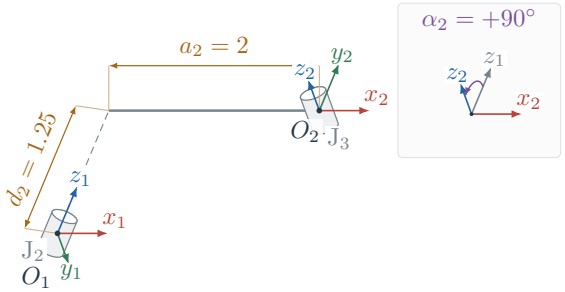
$d_1 = 1, \quad \theta_1 = q_1, \quad \alpha_1 = -60°, \quad a_1 = 0.8$



The full base height is 1 m. The 30° URDF roll of joint 2 gives $\alpha_1 = 30° - 90° = -60°$.

2 Shoulder to elbow: $\{1\} \to \{2\}$

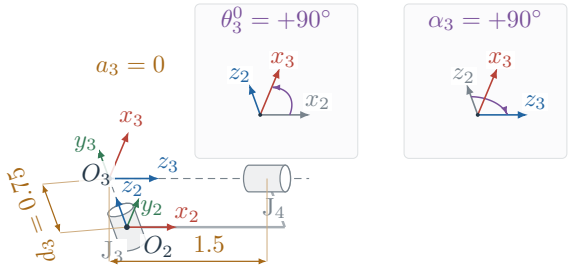
$d_2 = 1.25, \quad \theta_2 = q_2, \quad \alpha_2 = 90°, \quad a_2 = 2$



The x axes remain parallel. The offset along z_1 and the common normal along x_2 locate O_2 on joint 3.

3 Axis intersection: $\{2\} \to \{3\}$

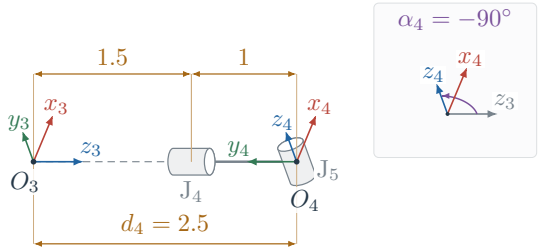
$d_3 = 0.75, \quad \theta_3 = q_3 + 90°, \quad \alpha_3 = 90°, \quad a_3 = 0$



O_3 is the intersection of the extended axes. Choosing $x_3 = y_2$ explains the fixed +90° offset in θ_3.

4 Reach the wrist: $\{3\} \to \{4\}$

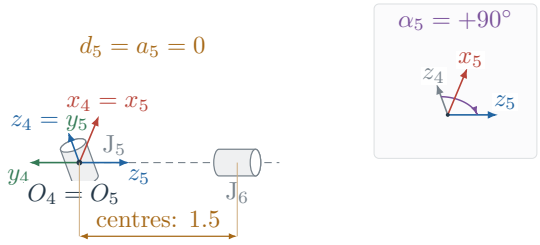
$d_4 = 2.5, \quad \theta_4 = q_4, \quad \alpha_4 = -90°, \quad a_4 = 0$



The distance starts at O_3, rather than the joint-4 cylinder. The two physical offsets add: $d_4 = 1.5 + 1$.

5 Coincident origins: $\{4\} \to \{5\}$

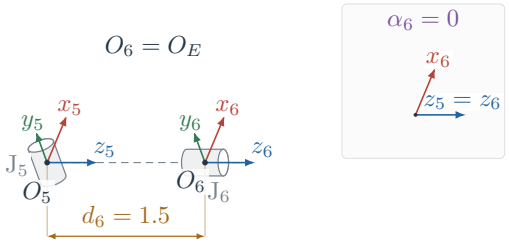
$d_5 = 0, \quad \theta_5 = q_5, \quad \alpha_5 = 90°, \quad a_5 = 0$



The origins coincide, although the joint centres do not. Here $x_4 = x_5$ and the extended joint-6 axis passes through the wrist centre.

6 Wrist to endpoint: $\{5\} \to \{6\}$

$d_6 = 1.5, \quad \theta_6 = q_6, \quad \alpha_6 = 0, \quad a_6 = 0$



At home the two frames have parallel axes. $O_6 = O_E$; the fixed endpoint alignment changes orientation only.

Figure 2: Standard DH frame assignment. Cylinders denote physical revolute joints J_i; dots mark DH origins O_i. x, y and z axes retain their colours. Dashed lines extend joint axes; gold dimensions locate the origins. Boxed angle sketches show the rotation $x_2 \to x_3$ about $+z_2$, giving θ_3^0, and the signed twist $z_{i-1} \to z_i$ about $+x_i$ (with the first direction translated to the same point). Cylinder orientation, size and viewing direction are reused between panels; drawings are schematic.